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2027 CYCLE

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CUET UG

Mathematics

Relations and Functions

Equivalence Relations, One-One/Onto, Composition, Inverse and Inverse Trigonometry · Section B1 I · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) 2026 syllabus — Mathematics/Applied Mathematics (319), the current 2027 baseline)	Skills the questions test
Types of relations	"Types of relations: Reflexive, symmetric, transitive and equivalence relations." (Section B1 Mathematics Unit I, part 1)	Running the three checks on a given relation and ruling equivalence in or out.
One-one and onto functions	"One to one and onto functions." (Section B1 Mathematics Unit I, part 1)	Classifying a function from its formula or its mapping picture.
Composition and inverse	"Composite functions, inverse of a function." (Section B1 Mathematics Unit I, part 1)	Evaluating $f \circ g$ in the right order; undoing a bijective function.
Inverse trigonometric functions	"Definition, range, domain, principal value branches. Graphs of inverse trigonometric functions." (Section B1 Mathematics Unit I, part 2)	Quoting domains, ranges and principal branches; using the sum identities.

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EXAM PATTERN

CUET UG pattern (NTA official information bulletin, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The 2026 syllabus houses this unit as **Section B1 Mathematics Unit I** — the language the rest of the paper speaks, since every later unit (calculus, algebra, probability) quietly assumes these definitions.

Relations — Pairs With Rules

A **relation** R from set A to set A is any collection of ordered pairs — formally a subset of the Cartesian product $A \times A$. "Is equal to", "is taller than", "differs from by 1" are all relations; the mathematics only records which pairs belong. Three properties classify them, a classic question type here is a six-pair relation with a verdict to deliver. Run the checks in order, because each one has a precise failure mode:

Property	The check	How it fails
Reflexive	$(a, a) \in R$ for **every** $a \in A$	One missing diagonal pair — scan all elements, no shortcuts

Property	The check	How it fails
Symmetric	$(a,b) \in R \Rightarrow (b,a) \in R$	A one-way arrow — some pair lacks its mirror
Transitive	$(a,b),(b,c) \in R \Rightarrow (a,c) \in R$	A chain that refuses to close — walk every 2-step route

An **equivalence relation** passes all three — it is the mathematics of "same category as": it sorts the set into disjoint classes where everything in a class relates to everything else. The identity relation $\Delta = \{(a,a)\}$ is always an equivalence relation (quick fact worth carrying), and "is congruent to" on triangles is the geometric flagship.

WORKED EXAMPLE — THE FULL THREE-CHECK AUDIT

$A = \{1,2,3\}$ and $R = \{(1,1),(2,2),(3,3),(1,2),(2,1)\}$. Is R an equivalence relation?

Solution. Reflexive: $(1,1),(2,2),(3,3)$ all present ✓. **Symmetric:** $(1,2)$ has its mirror $(2,1)$; the diagonal pairs mirror themselves ✓. **Transitive:** walk the chains — $1 \to 2$ then $2 \to 1$ needs $(1,1)$ ✓ present; $2 \to 1$ then $1 \to 2$ needs $(2,2)$ ✓; diagonals chain to themselves ✓. All three pass, so R is an **equivalence relation**. Now remove $(2,1)$ and re-run: reflexivity survives, symmetry dies ($(1,2)$ has no mirror), transitivity dies ($1 \to 2 \to 1$ had nowhere to land… the chain $1 \to 2$ then $2 \to 1$ no longer exists, but $1 \to 1$ then $1 \to 2$ still demands $(1,2)$ ✓ present — the honest kill is symmetry). One deleted pair, one verdict changed: this is exactly how option lists are built around a single pair.

Functions — One-One, Onto, Bijective

A **function** $f : A \to B$ assigns to **every** element of A exactly **one** partner in B. Two questions classify it. **One-one (injective):** do different inputs ever share an output? Algebraically, $f(x_1) = f(x_2)$ should force $x_1 = x_2$. **Onto (surjective):** does every element of the codomain get used — is the range equal to the codomain? A function that is both is **bijective**: a perfect one-to-one pairing of the two sets, with no leftovers on either side — and bijection is exactly the property an inverse function needs.

- **Linear functions** $f(x) = mx + c$ with $m \neq 0$ are bijective from $\mathbb{R} \to \mathbb{R}$ — the workhorses of inverse questions.
- $\boldsymbol{f(x) = x^2}$ **is not one-one on** $\mathbb{R}$ ($f(-2) = f(2)$) but is one-one on $[0, \infty)$ — domain choice changes the verdict, and MCQs lean on this.
- **Counting check:** from a finite set of n elements, a one-one function into itself must be onto as well — pigeonhole logic, and a quick MCQ evergreen.

WORKED EXAMPLE — CLASSIFY, THEN INVERT

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x + 3$. Show f is bijective and find f^{-1}.

Solution. One-one: $f(x_1) = f(x_2) \Rightarrow 2x_1 + 3 = 2x_2 + 3 \Rightarrow x_1 = x_2$ ✓. **Onto:** for any $y \in \mathbb{R}$, choose $x = \frac{y-3}{2}$; then $f(x) = y$ ✓ — every real number is hit. Bijective confirmed. **Inverse:** start from $y = 2x + 3$ and solve for x: $x = \frac{y-3}{2}$, so

$$f^{-1}(y) = \frac{y-3}{2} \quad \text{equivalently} \quad f^{-1}(x) = \frac{x-3}{2}.$$

Check the undo: $f^{-1}(f(5)) = f^{-1}(13) = \frac{13-3}{2} = 5$ ✓. The inverse **undoes** the machine — subtract the shift, divide the stretch, in reverse order of the build.

Composition and the Order Discipline

The **composite** $f \circ g$ means "run g first, then feed its output to f": $(f \circ g)(x) = f(g(x))$. Two disciplines matter. First, **order is content**: $f \circ g$ and $g \circ f$ are different machines in general, and the exam tests the difference directly. Second, **compatibility**: g's outputs must land inside f's domain, or the composite does not even exist.

WORKED EXAMPLE — SAME PARTS, DIFFERENT ORDER

$f(x) = 2x + 3$ and $g(x) = x^2$. Find $(f \circ g)(x)$ and $(g \circ f)(x)$, and evaluate both at $x = 3$.

Solution. Inside first:

$$(f \circ g)(x) = f(g(x)) = 2x^2 + 3, \quad (g \circ f)(x) = g(2x + 3) = (2x + 3)^2.$$

At $x = 3$: $(f \circ g)(3) = 2 \cdot 9 + 3 = 21$, while $(g \circ f)(3) = 9^2 = 81$. Same two functions, wildly different results — the order is the whole question. One-line check for MCQs: $f \circ g$ wears g's inside clothes, so square first here.

Inverse Trigonometric Functions — the Principal Branch Idea

Sine repeats its values forever, so $\sin : \mathbb{R} \to [-1, 1]$ is hopelessly **not one-one** — no inverse function can exist on the whole line. The fix is elegant: **restrict the domain to one clean branch** where the function is bijective, then invert. The chosen branch is the **principal value branch**, and the six pairs of domains and ranges are the single most quotable table in the unit:

Function	Domain	Principal range	One-line character
$\sin^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	odd function; the "steep S"
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$	the downward half-story
$\tan^{-1} x$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$	open ends — the bounds are never reached
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$	tan's partner across the sum rule

Function	Domain	Principal range	One-line character
$\sec^{-1} x$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \{\frac{\pi}{2}\}$	lives only outside the unit interval
$\operatorname{cosec}^{-1} x$	$\mathbb{R} - (-1, 1)$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$	skips zero where it would explode

- **Why the funny brackets?** $\tan x$ never equals $\pm\frac{\pi}{2}$ (it runs to infinity there), so $\tan^{-1}$ can never **return** those values — open interval. Similarly sec and cosec skip $\frac{\pi}{2}$ and 0 where they would be undefined.
- **The sum identities** fall straight out of the branches: $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ (their ranges stack to a half-turn), and $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ likewise. These two lines answer a surprising number of MCQs alone.
- **Graphs carry the same message:** $\sin^{-1}$ is the sine wave stood upright on the restricted branch; $\tan^{-1}$ is the S-curve flattening toward its open bounds $\pm\frac{\pi}{2}$ — the horizontal asymptotes **are** the principal range.

WORKED EXAMPLE — BRANCHES AT WORK

Evaluate $\sin^{-1}\left(\frac{1}{2}\right) + \cos^{-1}\left(\frac{1}{2}\right)$ and $\tan^{-1}(1) + \cot^{-1}(1)$ without a calculator.

Solution. By the sum identity, $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ for every x in $[-1, 1]$ — so the first sum is $\frac{\pi}{2}$ with no individual evaluation needed. Same for the second: $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$. Both answers are $\frac{\pi}{2}$. (Cross-check the first the slow way: $\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$ because $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$, and $\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$; $\frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{2}$ ✓ — the identity and the arithmetic agree.)

EVERYDAY EXAMPLE — THE LIFT WITH ONE BUTTON PER FLOOR

A building's lift panel is a **function**: press a floor, get taken there. If two buttons lit the same floor, the panel would be ambiguous — not one-one. If some floor had no button, some destinations would be unreachable — not onto. Only a panel with exactly one button per floor, every floor covered, can be **undone**: you can always send the lift back. Restricting sine's domain to its principal branch is exactly deleting the duplicate buttons — keeping one clean tower section where every floor appears once, so the return trip (the inverse) becomes possible.

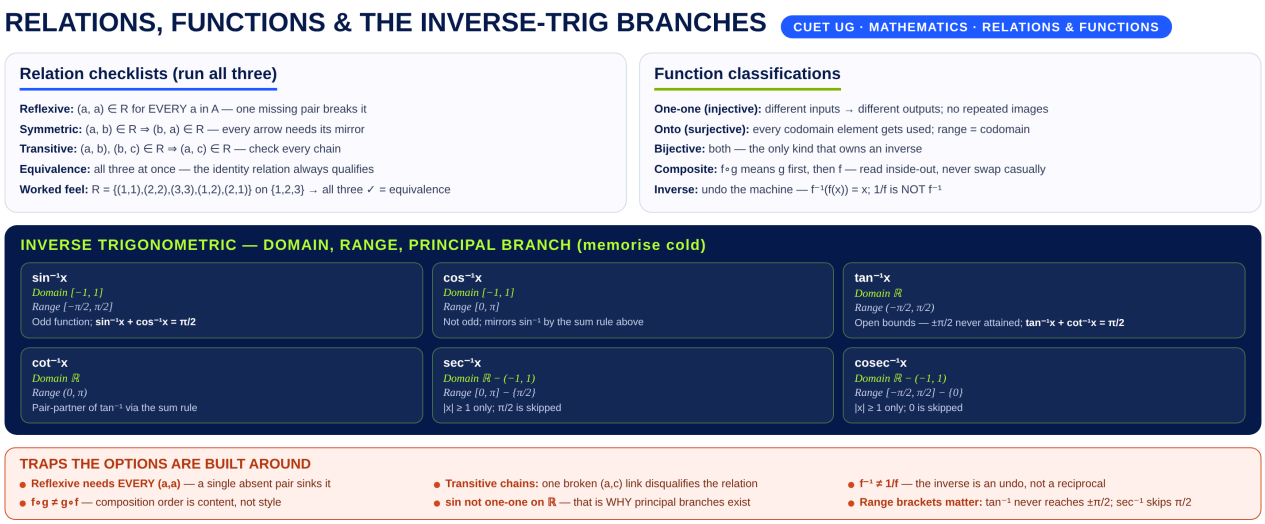


Figure 1: The unit on one board: the three relation checks, the function classifications, the six principal branches, and the traps between them.

Formula Cheatsheet — the Whole Chapter on One Page

FORMULA CHEATSHEET — RELATIONS AND FUNCTIONS

Formula / rule	Letters mean	Fire it when
Reflexive: $(a, a) \in R$ for every a	no exceptions	relation audits
Symmetric: $(a, b) \in R \Rightarrow (b, a) \in R$	mirror pairs	relation audits
Transitive: $(a, b), (b, c) \Rightarrow (a, c)$	close every chain	relation audits
Equivalence = all three	sorts into classes	verdict questions
One-one: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$	no shared images	classification
Onto: range = codomain	nothing unused	classification
Bijective = one-one + onto	the inverse license	inverse prep
$(f \circ g)(x) = f(g(x))$	inside first	composites
Inverse: solve $y = f(x)$ for x	undo the machine	bijective functions

Formula / rule	Letters mean	Fire it when
Linear $mx + c$, $m \neq 0$: bijective $\mathbb{R} \to \mathbb{R}$	workhorse	quick classifications
$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$	branch stacking	no-calculator sums
$\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$	same trick	no-calculator sums
Branches: $\sin^{-1}[-1,1] \to [-\frac{\pi}{2}, \frac{\pi}{2}]$; $\cos^{-1} \to [0, \pi]$; $\tan^{-1} \mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$	domain-range pairs	table recall

Common Errors and Special Cases

COMMON ERRORS AND SPECIAL CASES — RELATIONS AND FUNCTIONS

The mistake	What is right	The trap it sets
Reflexive checked on listed pairs only	Every element of A needs its (a, a)	Relations missing one diagonal pair
Transitivity skipped when it "looks fine"	Walk every 2-step chain	Relations with one unclosed chain
Writing $f^{-1} = \frac{1}{f}$	The inverse undoes; the reciprocal inverts values	The oldest wrong option in the bank
Swapping $f \circ g$ and $g \circ f$	Order changes the machine	Same-parts-different-answer options
Inverse attempted on non-bijective f	Bijection is the license	x^2 "inverse" candidates
Declaring x^2 never invertible	One-one on restricted domains	Domain-choice verdict flips
Square brackets on $\tan^{-1}$ range	$(-\frac{\pi}{2}, \frac{\pi}{2})$ — bounds unreachable	Bracket-flipped range options
Forgetting $\sec^{-1}$, $\operatorname{cosec}^{-1}$ domains	Outside the unit interval only	Whole-realinverse domain options

Sixty-Second Self-Test

Statement	Verdict
Every equivalence relation is reflexive, symmetric and transitive.	True — by definition
$f(x) = x^2$ is one-one on $\mathbb{R}$.	False — $f(-2) = f(2)$
$f^{-1}(x) = \frac{1}{x}$ for $f(x) = 2x$.	False — $f^{-1}(x) = \frac{x}{2}$
The range of $\tan^{-1}$ includes $\frac{\pi}{2}$.	False — open bound, never attained
$\sin^{-1}\left(\frac{1}{2}\right) + \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{2}$.	True — the sum identity
Composition order does not matter.	False — $f \circ g \neq g \circ f$ in general

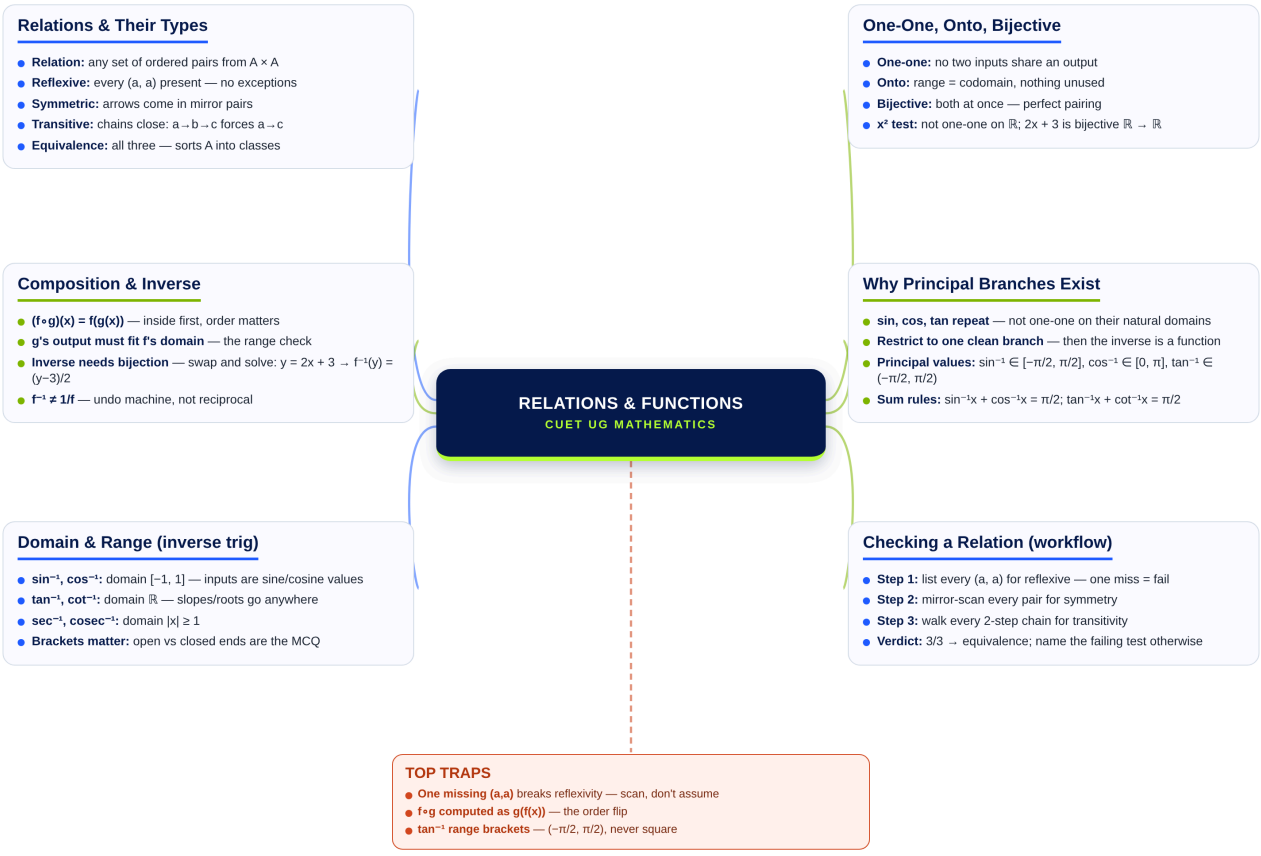


Figure 2: The chapter in one look: relation checks, function classes, composition discipline, and the six principal branches.

Practice Questions — CUET UG Pattern (10 MCQs)

Ten questions in the CUET UG style. Each correct answer earns +5; each incorrect one costs −1. Definitions decide most of this set — run the checklists, not the vibes.

Q1. On $A = \{1, 2, 3\}$, the relation $R = \{(1, 1), (2, 2), (3, 3), (2, 3), (3, 2)\}$ is:

(A) reflexive and symmetric but not transitive
(B) symmetric and transitive but not reflexive
(C) reflexive and transitive but not symmetric
(D) an equivalence relation

Q2. The function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 3x - 4$ is:

(A) one-one but not onto
(B) onto but not one-one
(C) neither one-one nor onto
(D) bijective

Q3. If $f(x) = 2x + 3$ and $g(x) = x^2$, then $(f \circ g)(x)$ equals:

(A) $(2x + 3)^2$
(B) $2x^2 + 3$
(C) $4x^2 + 9$
(D) $2x^2 + 9$

Q4. The inverse of $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 5x - 7$, is:

(A) $\frac{1}{5x-7}$
(B) $\frac{x-7}{5}$
(C) $5x + 7$
(D) $\frac{x+7}{5}$

Q5. The domain of $\sin^{-1} x$ is:

(A) $\mathbb{R}$
(B) $(-1, 1)$
(C) $[-1, 1]$
(D) $[-\frac{\pi}{2}, \frac{\pi}{2}]$

Q6. The principal value branch of $\cos^{-1} x$ is:

(A) $(-\frac{\pi}{2}, \frac{\pi}{2})$
(B) $[0, \pi]$
(C) $[-\frac{\pi}{2}, \frac{\pi}{2}]$
(D) $(0, \pi)$

Q7. $\sin^{-1} x + \cos^{-1} x$ equals, for $x \in [-1, 1]$:

(A) 0
(B) pi
(C) $\frac{\pi}{2}$
(D) depends on x

Q8. Which function is NOT one-one on its stated domain?

(A) $f(x) = 2x$ on $\mathbb{R}$
(B) $f(x) = x^3$ on $\mathbb{R}$
(C) $f(x) = x^2$ on $\mathbb{R}$
(D) $f(x) = x + 5$ on $\mathbb{R}$

Q9. The relation "is equal to" on any set is:

(A) an equivalence relation
(B) reflexive only
(C) symmetric only
(D) transitive only

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Q10. $\tan^{-1}(1) + \cot^{-1}(1)$ equals:

(A) $\frac{\pi}{4}$ **(B)** $\frac{\pi}{2}$
(C) 1 **(D)** π

Detailed Solutions

Sol. Q1. Reflexive: all three diagonal pairs present ✓. Symmetric: $(2,3)$ and $(3,2)$ mirror ✓. Transitive: $2 \to 3$ then $3 \to 2$ demands $(2,2)$ ✓ present… and $2 \to 3$, $3 \to 2$ chain closes; but check $2 \to 3$ then $3 \to 3$ needs $(2,3)$ ✓. The silent failure: $3 \to 2$ then $2 \to 3$ demands $(3,3)$ ✓ present. Wait — all chains close; re-verify: no chain produces a missing pair, so it IS transitive and thus an equivalence relation. The audit wins again.

Answer: (A)

Sol. Q2. Slope 3 is non-zero: one-one. For any y, $x = \frac{y+4}{3}$ hits it: onto. Bijective — the linear workhorse, with inverse $\frac{y+4}{3}$ ready.

Answer: (D)

Sol. Q3. Inside first: $g(x) = x^2$, then f doubles and shifts: $2x^2 + 3$. Option (A) is the swapped order $(g \circ f)$ — the order trap with numbers on.

Answer: (B)

Sol. Q4. Solve $y = 5x - 7$ for x: $x = \frac{y+7}{5}$. So $f^{-1}(x) = \frac{x+7}{5}$. Option (A) is the reciprocal lure; option (B) flipped the sign — undo means add the 7 back first.

Answer: (D)

Sol. Q5. The inverse sine accepts exactly the values sine can output: $[-1, 1]$, closed at both ends since sin attains ± 1. Option (D) quotes a **range**, not a domain.

Answer: (C)

Sol. Q6. The principal branch of cosine spans $[0, \pi]$ — closed both ends, monotone decreasing, bijective onto the range. Option (A) belongs to $\tan^{-1}$-family thinking.

Answer: (B)

Sol. Q7. Branch stacking: $\sin^{-1}$ covers $[-\frac{\pi}{2}, \frac{\pi}{2}]$ and $\cos^{-1}$ covers $[0, \pi]$; for the same x the two angles sit symmetric about $\frac{\pi}{4}$ and always sum to $\frac{\pi}{2}$ — at every x in $[-1, 1]$.

Answer: (C)

Sol. Q8. On all of $\mathbb{R}$, x^2 repeats images: $f(-3) = f(3) = 9$. The others are strictly increasing (or decreasing) — one-one by monotonicity. Domain choice is the whole question.

Answer: (C)

Sol. Q9. "Is equal to" passes all three: $a = a$ ✓ reflexive; equality is two-way ✓ symmetric; $a = b$, $b = c$ force $a = c$ ✓ transitive. Equivalence — the flagship example.

Answer: (A)

Sol. Q10. The sum identity fires: $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ for every real x — no individual evaluation needed. (Slow check: both are $\frac{\pi}{4}$, summing to $\frac{\pi}{2}$ ✓.)

Answer: (B)

Wrong-Answer Clinic

If you picked this	The thinking slip	The lasting fix
(A) in Q2	Assumed linear = one-one only	Check onto too: every y has an $\frac{y+4}{3}$ — both boxes tick
(A) in Q3	Composed in the written-order reverse	(f compose g) = f after g: g's clothes inside f's frame
(A) in Q4	Confused inverse with reciprocal	Inverse = solve-and-swap; reciprocal = flip the value
(D) in Q5	Quoted the range as the domain	Domain = what feeds it (sine's outputs); range = what it returns
(A) in Q6	Mixed branches across functions	One table, six rows — memorise as pairs, not vibes
(A) in Q7	Started evaluating each term	Sum identities skip the work — try them before any calculator

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